

Vector Fields Behind Jujutsu Kaisen: Gojo Satoru's Blue, Red and Purple



Amirlan Kalmukhan

Essays on the Science of Fiction, No. 2

September 2026

1 Introduction

When Gojo Satoru explains his two strongest attacks, he does not talk about force. He talks about arithmetic.

Describing Blue to an opponent in chapter 69, he calls it bringing the concepts of negative numbers and negative distance into reality, and then letting real space rush in to fill the impossibility. Red, which he could not use for years afterwards, runs on reverse cursed technique, and chapter 74 defines that precisely: you multiply a negative by another negative to get a positive. Purple happens when the two collide, and the official fanbook says the collision produces an imaginary mass.

Negative, positive, imaginary. Those are not three moods. They are two specific places on the number line and one place that is not on it at all.

Akutami went further than the dialogue, into the names. Blue is 術式順転, *jutsushiki juten*, the technique run forward. Red is 術式反転, *jutsushiki hanten*, the technique reversed. Purple is 虚式, *kyoshiki*, and 虚 is the character Japanese mathematics uses for imaginary numbers: 虚数, *kyosū*, sits opposite 実数, *jissū*, the real numbers. Forward, reversed, imaginary. The English translations, Lapse and Reversal and Hollow, hide a symmetry that the Japanese states outright.

This essay takes the naming seriously and reaches for the branch of mathematics built to handle exactly what Gojo is doing. Vector calculus was invented in the nineteenth century to answer one question about any field filling space: is stuff flowing into this point, or out of it? There is a single number that answers it, and the entire Limitless offence turns out to be a fight over its sign.

Three results, in the order they surprised me. Purple has exactly zero net outward force, which takes one line to prove and explains why it erases things instead of pushing them. Combining a divergent field with a convergent one multiplicatively always yields an imaginary quantity, not merely for Gojo's particular numbers but for every such pair. And the four kilometres of Tokyo that Purple carved between Shibuya and Shinjuku are the wrong shape for an explosion by a factor of about eighty, which rules out the reading almost everyone reaches for first.

2 What the Three Techniques Actually Do

The Limitless (無下限呪術, *Mukagen Jujutsu*) is the inherited technique of the Gojo clan, and the manga is unusually direct about its mathematics. Chapter 69 states that it operates the same way convergent and divergent sequences do. Its neutral state, Infinity, takes the finite distance between Gojo and anything approaching him and divides it endlessly, so that an attack slows without ever arriving.



Figure 1: Red and Blue held apart, one in each hand. This is the whole essay in a single frame: on the left a source, on the right a sink, and the only thing separating them is a distance Gojo controls. Everything that follows is about what that distance is worth. Image source: Jujutsu Kaisen Wiki (Fandom), *Limitless*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

Blue (術式順転「蒼」) first appears in chapter 52. Canon describes it as infinitely converging space to a single point to create a void-like effect, compressing the space that is there and compelling everything in range to fill the void that results. The size of the field scales with how much cursed energy Gojo pours in. A direct hit does not knock a person away. Chapter 52 has the target implode.

Red (術式反転「赫」) appears earlier, in chapter 14, and is much harder. It runs on positive energy from reverse cursed technique rather than on cursed energy, and canon says it represents the concept of divergence brought into reality, producing repulsion where Blue produced attraction. The figure that matters for this essay is in chapter 84: Red's output is at least twice that of Blue. Gojo, a prodigy with the Six Eyes, could not perform it as a second year student and only managed it after nearly dying.

Purple (虚式「茈」) collides the two. The fanbook puts it plainly: the repulsive force of Red combined with the attractive force of Blue brings forth an imaginary mass, which is then pushed at the target. It has erased a large portion of a man's left side (chapter 75), annihilated a stretch of forest (chapter 52), and torn through roughly four kilometres of buildings between Shibuya and Shinjuku (chapter 223). Chapter 235 adds a detail worth holding onto: left completely unchecked, Purple can injure the person using it.

3 Divergence, the Number That Tells a Source From a Sink

Imagine a small box drawn in the middle of a flowing river. Water crosses its faces, some going in, some coming out. Add up everything entering, subtract everything leaving, and you get one number for that box. Physicists call it the flux.

For an ordinary patch of river the number is zero. Whatever flows in flows out again, because water is not being created or destroyed inside the box. But put a drain in the box and more goes in than comes out. Put a hose in it and more comes out than went in. The flux stops being zero, and its sign tells you which.

Now shrink the box down to a point and divide by its volume as you go. The number you are left with is the divergence of the field at that point, written $\nabla \cdot \mathbf{F}$. It is positive at a source, negative at a sink, and zero everywhere ordinary.

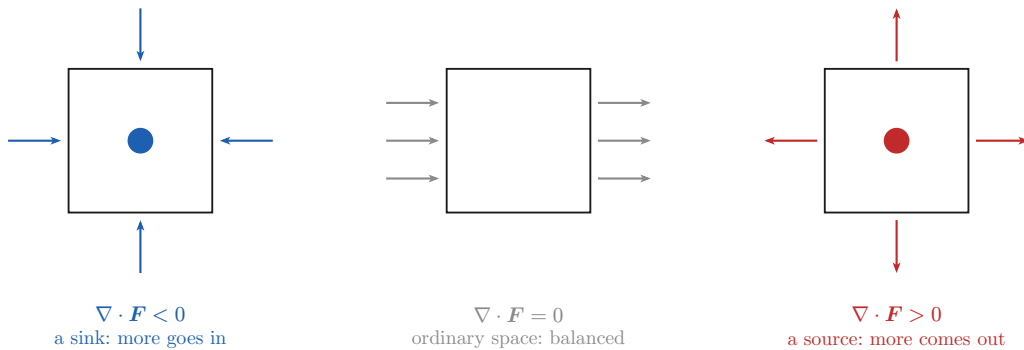


Figure 2: Divergence is bookkeeping. Draw a box, count what crosses its walls, and subtract. If the total is negative the box contains a sink and space is being consumed inside it; if positive it contains a source. Shrink the box to a point and the number you are left with is $\nabla \cdot \mathbf{F}$, defined at every point in space. Gojo's entire offence is a fight over this one sign.

The reason this is more than bookkeeping is a theorem Carl Friedrich Gauss published in the 1810s. The divergence theorem says that the flux through any closed surface equals the total divergence contained inside it:

$$\oint_S \mathbf{F} \cdot d\mathbf{A} = \int_V (\nabla \cdot \mathbf{F}) dV.$$

Read the consequence carefully, because the rest of this essay hangs on it. The flux depends only on what is inside the surface. Not on the surface's shape, not on its size, not on how the sources inside are arranged. Two configurations with the same total enclosed divergence push on the outside world identically, however different they look.

4 Blue, and the Arithmetic of Negative Distance

Blue is a sink. Gojo makes a point where space is being consumed, and the divergence around it is negative.

His own description is sharper than “attraction,” and the difference is worth slowing down for. A force of attraction is something a target feels and responds to. Negative

distance is not a force at all. It is an inconsistency. If a region contains less space than it ought to, the surrounding space is not tempted to move in, it is obliged to, because the alternative is a gap, and a gap is not a state space is allowed to be in.

That reading predicts the right symptom. If Blue pulled, its victims would accelerate toward the orb and arrive at it. Canon says they implode, collapsing inward on themselves where they stand. They are not travelling toward anything. They are being used as filler.

For a sink of strength Q the field falls off as the inverse square, and the divergence theorem gives the flux through any surface around it:

$$\oint_S \mathbf{F} \cdot d\mathbf{A} = -Q.$$

The minus sign is Blue. Everything else about the technique, the orb, the range scaling with cursed energy, the buildings folding inward, is that minus sign made large.

5 Red, and Why Reversal Costs Two Negatives



Figure 3: Red. Canon calls it divergence brought into reality and puts its output at no less than twice Blue's. That factor of two is the one number in this essay that needs an argument rather than a calculation. Image source: Jujutsu Kaisen Wiki (Fandom), *Cursed Technique Reversal: Red*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

Red is the same field with the sign reversed. The divergence is positive, the flux through any enclosing surface is $+Q$, and space is produced at the centre rather than consumed. So far this is symmetric, and that symmetry is a problem, because canon says the two techniques are not equal. Red's output is at least double Blue's.

Flipping a sign does not double anything. The distance from zero to -1 and the distance from zero to $+1$ are the same distance. So either the manga's number is decoration, or Red is not simply Blue with a minus in front.

I think it is the second, and the reason sits in the names. Blue is 順転, forward: Gojo channels cursed energy into the Limitless and runs it the normal direction. One

input, one deficit. Red is 反転, reversal, and reversal in this world is not a sign flip. It is reverse cursed technique, which chapter 74 defines exactly: you multiply a negative by another negative to obtain a positive. That takes two negatives. Gojo has to produce and combine twice the input before Red exists at all, and the manga is consistent that a Limitless technique's output scales with what is poured into it.

So Red is not Blue negated. Red is Blue's input consumed twice over and squared into positivity, and "at least twice" is what you get if output tracks the number of negatives spent.

There is a competing reading worth naming. If you measure Red's displacement not from neutral but from Blue, then moving from -1 to $+1$ covers two units rather than one, and you get the factor that way. It is tidier. I prefer the first because it explains something the second cannot, which is why Red was so much harder to learn. Travelling twice as far along an axis is not difficult. Manufacturing a second negative and multiplying is, and canon calls reverse cursed technique rare even inside the three great families. Gojo, born with the Six Eyes, needed a near-death experience to manage it.

Either way, the honest status of this section is a reading that reproduces a stated number, not a derivation that predicts one.

6 Purple, a Field With No Outside

Now collide them. Put Blue's sink at one point and Red's source a short distance away, and ask what the combination looks like from outside.

Gauss answers in a line. Take any closed surface containing both. The flux through it equals the total divergence enclosed, and the total divergence enclosed is

$$(+Q) + (-Q) = 0.$$

Zero. Not small, not mostly cancelled. Exactly zero, through every surface, at every distance, for as long as both halves are present.

A field with no net flux exerts no net push and no net pull on anything around it. Purple cannot blow you away and cannot draw you in. That is not a weakness of the technique, it is an identity, and it holds no matter how much cursed energy Gojo spends.

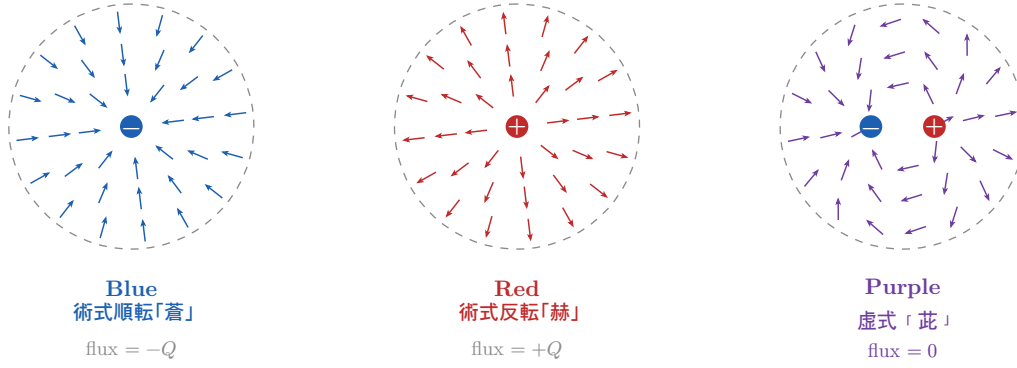


Figure 4: The three techniques as vector fields, with a dashed surface drawn around each. Blue consumes space, so every arrow points inward and the flux through the surface is $-Q$. Red produces it and the flux is $+Q$. Purple holds both at once, and the flux through the same surface is exactly zero: outside the pair, the two halves cancel completely. Notice what the arrows do in the third panel. Nothing leaves, nothing enters, and everything happens in the gap.

Which leaves an obvious problem. Purple is the most destructive thing in the series. How does a field with no net force flatten four kilometres of Tokyo?

Because the outside cancelling says nothing about the inside. The energy stored in a source and sink separated by a distance r goes as

$$U \propto \frac{Q^2}{r},$$

and that quantity does not care that the monopoles cancel at infinity. As Gojo closes the gap, U grows without bound. Halve the separation and you double the energy. Take r to zero and there is no limit to take.

So Purple's structure is: nothing at all on the outside, and an unbounded amount in the middle. Canon's warning in chapter 235 that an unchecked Purple can injure Gojo himself is what holding a divergent quantity should feel like.

There is a word for a shape that is empty everywhere except inside. 虚. Hollow.

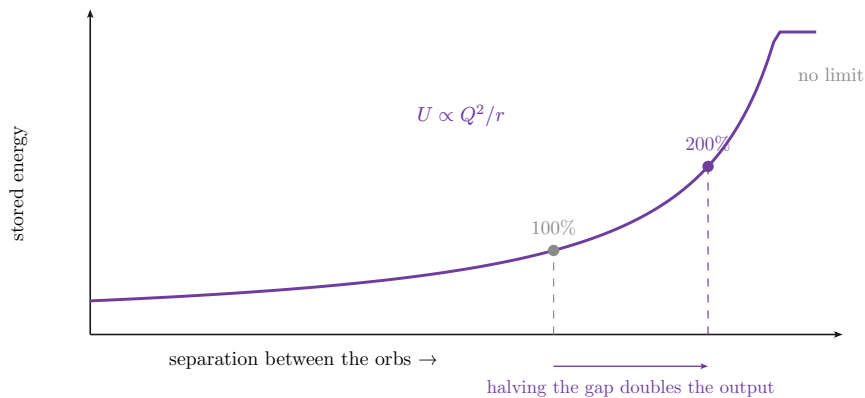


Figure 5: Energy stored between the two orbs as Gojo closes the gap. The far field cancels exactly, so none of this is visible from outside until it is released. The curve has no ceiling: every halving of the separation doubles the output. Chapter 223 has Gojo add the full hand signs and incantation he normally omits and reach 200 percent, which on this curve is not extra fuel at all. It is one more halving of the gap, bought with precision.

Now the sign. Blue and Red are both scalings of the same space, and scalings compose by multiplication rather than addition: applying one and then the other multiplies their factors. The single scaling equivalent to two others is their geometric mean, $\sqrt{ab}$. With Blue at -1 and Red at $+2$:

$$\sqrt{(+2)(-1)} = \sqrt{-2} = i\sqrt{2}.$$

The result is imaginary, and not because of anything special about Gojo's numbers. For any positive a and any negative b , $\sqrt{ab}$ is purely imaginary, every single time, because no real number squared is ever negative. Combine a source with a sink multiplicatively and you leave the real line. You have no choice in the matter.

That is the sense in which Purple was inevitable from the moment Gojo could perform both halves. It is also why adding them instead gives nothing: $(+1) + (-1) = 0$, the neutral Infinity Gojo already had.

7 What an Imaginary Mass Actually Does

The fanbook's phrase is imaginary mass, and physics has a standard place where imaginary mass turns up.

Start with the relativistic energy of a particle of rest mass m_0 moving at speed v :

$$E = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}}.$$

Put in an imaginary rest mass, $m_0 = i\mu$. For E to come out real, the square root underneath has to be imaginary too, which requires $v > c$. Gerald Feinberg named these hypothetical particles tachyons in 1967, and they behave backwards from everything familiar:

$$E = \frac{\mu c^2}{\sqrt{v^2/c^2 - 1}}.$$

As v falls toward c from above, E runs to infinity. As v climbs, E falls toward zero. A tachyon speeds up when you take energy out of it, and slows toward light speed as you pour energy in.

So one reading of Purple is a faster-than-light projectile. It is also the reading physics abandoned, and the correction is more interesting than the original.

Aharonov, Komar and Susskind showed in 1969 that a field with imaginary mass sends no faster-than-light signals. What $m^2 < 0$ really means is that you are sitting at a maximum of the potential instead of a minimum. The state is unstable. It rolls off.

This is not an obscure corner of the subject, it is the Higgs mechanism. Write the potential

$$V(\varphi) = -\mu^2|\varphi|^2 + \lambda|\varphi|^4,$$

and the minus sign in front of μ^2 is precisely an imaginary mass term. It is the reason the symmetry breaks: the origin is a hilltop rather than a valley, so the field cannot stay there and slides out to the brim at $|\varphi| = \mu/\sqrt{2\lambda}$. Imaginary mass does not mean “moves faster than light.” It means “this is not a place anything can remain.”

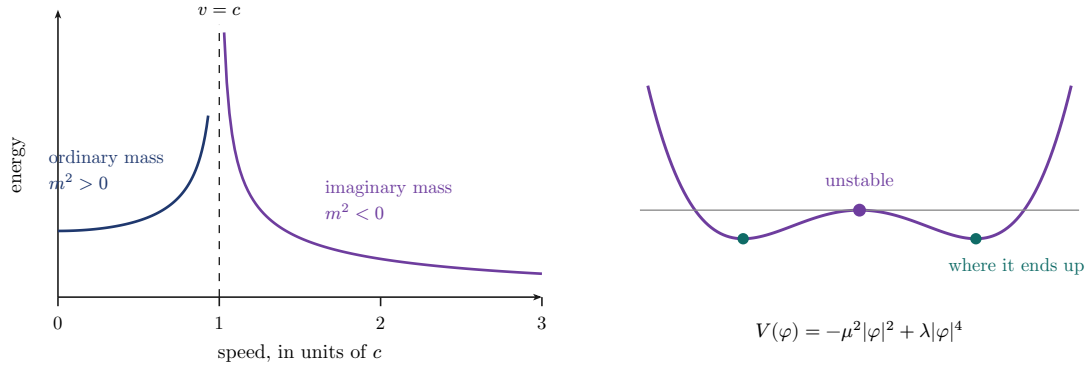


Figure 6: Left: energy against speed. An ordinary particle (navy) needs infinite energy to reach light speed from below. A particle of imaginary mass (purple) lives on the far side and runs the other way, gaining speed as it loses energy. Right: what imaginary mass means in field theory. The negative μ^2 term turns the origin from a valley into a hilltop, so nothing can sit there. This is the Higgs potential, and the same sign that makes the Standard Model work is the sign the fanbook attaches to Purple.



Figure 7: Purple, fired at Toji Fushiguro in chapter 75. What it did to him is worth stating in the manga’s own words: it erased a large portion of the left side of his body. Not crushed, not blasted off. Erased. Image source: Jujutsu Kaisen Wiki (Fandom), *Hollow Technique: Purple*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

Apply that to Purple and the behaviour falls straight out. An imaginary mass does not deliver momentum to a target. It marks a region where the present state has stopped being stable, and everything in that region has to become something else.

Which is what the manga draws, consistently, in language nobody seems to have picked up on. Toji is not knocked back by Purple; a large portion of his left side is

erased. The forest is not flattened; it is annihilated. The buildings between Shibuya and Shinjuku are not blown over; they are eviscerated. Every single description is removal rather than displacement. Displacement is what forces do, and we have already proved Purple has no net force to displace anything with.

8 Four Kilometres in a Straight Line



Figure 8: Purple arriving in Shinjuku, chapter 223. Buildings are standing on both sides of the channel. That detail does more work than it looks like it does. Image source: Jujutsu Kaisen Wiki (Fandom), *Hollow Technique: Purple*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

Chapter 223 gives the one number big enough to test against the real world. Gojo fires Purple from Shibuya at Sukuna waiting in Shinjuku, and it eviscerates around four kilometres of buildings between them.

Shibuya Station to Shinjuku Station is 3.51 kilometres in a straight line. Akutami’s geography is sound, which is a good sign for taking the rest of the figure seriously.

Now test the reading everyone reaches for first: that Purple is an explosion, a very large bomb delivered at range. Explosions scale by the cube root of yield, so one comparison sets the scale for all of them. Hiroshima was roughly 15 kilotons and produced near-total destruction out to about 1.6 kilometres. Scaling up:

$$Y = \left(\frac{4.0}{0.649} \right)^3 \approx 234 \text{ kilotons},$$

about sixteen Hiroshimas. That is a lot, but it is not absurd for this series, and if the argument stopped at energy the explosion model would survive.

It does not stop at energy, because energy was never the problem. Shape is. A 234 kiloton detonation in Shibuya destroys a disc of radius four kilometres, which is $\pi \times 16 \approx 50$ square kilometres of central Tokyo, in every direction at once, including the

direction Gojo was standing in. What chapter 223 shows is a corridor: four kilometres long and, measured off the art, somewhere between one and two hundred metres wide. Call it 0.6 square kilometres.

The two differ by a factor of about eighty. And unlike energy, you cannot tune your way out of it. An explosion is isotropic, so it makes a disc, and no choice of yield turns a disc into a line.

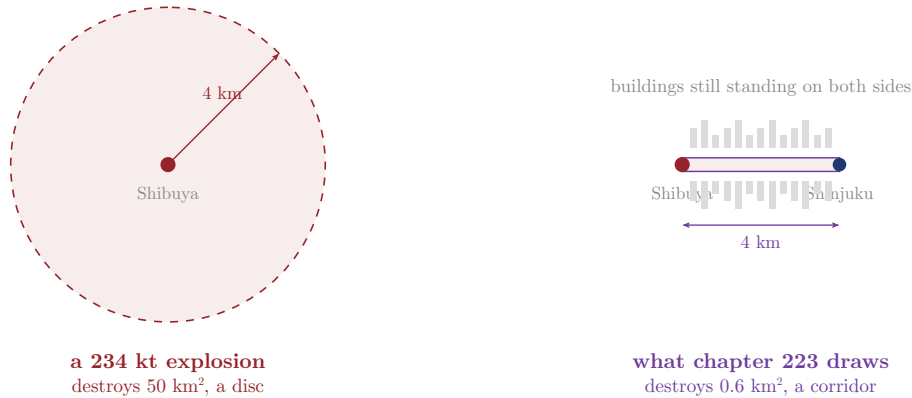


Figure 9: The same four kilometres, drawn two ways. On the left, the disc an explosion large enough to reach that far would flatten, with Gojo at the centre of it. On the right, the channel the manga actually shows, with the city intact a hundred metres to either side. The mismatch is roughly eighty to one, and no amount of extra yield fixes it, because the disagreement is about geometry rather than power. A narrow track of destruction with everything standing on both sides is the path of something that travelled and converted what it met. It is not a blast radius.

That is the strongest evidence in the essay for the unstable-vacuum reading over the projectile one, and it comes from a drawing rather than from dialogue. A bubble of instability expanding at close to light speed along a line, converting whatever it passes through and leaving its flanks untouched, produces exactly this picture. A bomb cannot.

9 Conclusion

Gojo's three Limitless attacks are three points and one operation.

Blue is a sink. The divergence around it is negative, the flux through any surrounding surface is $-Q$, and its victims implode rather than travel because negative distance is an inconsistency that space is obliged to repair rather than a force anything responds to. Red is the same field with positive divergence, and its output is at least double Blue's because reversal in this world is not a sign flip but a multiplication requiring two negatives, which is also why it took Gojo a near-death experience to learn. Purple is both at once, and three things follow with no freedom left in them. The net flux is exactly zero, so Purple has no push and no pull. The stored energy diverges as the gap closes, so its output has no ceiling and the canonical 200 percent is one halving of the separation rather than extra fuel. And the geometric mean of a positive and a negative is always imaginary, so Purple's amplitude leaves the real line the moment both halves exist.

An imaginary amplitude does not describe a fast projectile. In every serious treatment since 1969 it describes a region where the current state is unstable and must decay, which is why Purple erases rather than pushes, and why four kilometres of Tokyo came out as a corridor instead of the fifty square kilometre disc an explosion of that reach would have left.

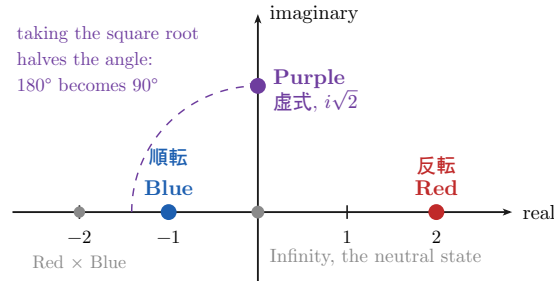


Figure 10: The three techniques on the complex plane. Blue sits at -1 , neutral Infinity at the origin, Red at $+2$. Their product lands at -2 , a hundred and eighty degrees round from the positive axis, and taking a square root halves that angle. Ninety degrees is straight up, off the real line entirely. Attraction and repulsion are the two directions available along the horizontal axis, and Purple is the technique that does not lie on it.

There are limits here and they should be said out loud rather than left for someone else to notice. Cursed energy obeys no conservation law anyone has written down, and nothing above constrains how much of it Gojo has. The decision to combine Blue and Red multiplicatively rather than additively is a modelling choice, and it is the choice the whole imaginary result rests on; add them instead and you get zero, which is the neutral Infinity he already had and no essay. The corridor width is eyeballed off a drawing, so the factor of eighty is good to about a factor of two. And the reading of the canonical doubling explains that number without deriving it, which is a weaker thing than it sounds like when stated confidently.

What survives all of that is the part that never touched a number. Any sink and any source of equal strength enclose zero net divergence, so their combination cannot push or pull, whatever they are made of. Any positive quantity multiplied by any negative one has an imaginary square root, whatever the magnitudes. Those two facts are the whole of Purple, and neither of them needed the manga's permission.

One last thing, and it is the detail I have not been able to put down. 虚式 is usually rendered Hollow Technique, and 虚 does mean hollow: empty, void, the character in 虚空, empty space. It is also, in every Japanese mathematics textbook ever printed, the character for imaginary. 虚数, *kyosū*, imaginary number, set against 実数, *jissū*, the real ones.

Both readings are true at once, and they are describing the same object from two sides. Purple's field is hollow in the strict sense: zero flux through every surface you can draw around it, nothing whatsoever on the outside. And Purple's amplitude is imaginary in the strict sense: perpendicular to the axis where attraction and repulsion are the only two things on offer.

Blue goes one way along that axis and Red goes the other, and between them they use up everything a force is permitted to do. Purple is the attack that steps off the line. Gojo’s third technique is not stronger than his first two. It is somewhere else.

References

- [1] G. Akutami, *Jujutsu Kaisen*, Shueisha, 2018–2024. Chapters 2, 14, 52, 69, 74, 75, 84, 223 and 235.
- [2] G. Akutami, *Jujutsu Kaisen Official Fanbook*, Shueisha, 2021, p. 158.
- [3] C. F. Gauss, “Theoria attractionis corporum sphaeroidicorum ellipticorum homogeneorum,” *Commentationes Societatis Regiae Scientiarum Gottingensis Recentiores*, vol. 2, 1813.
- [4] H. M. Schey, *Div, Grad, Curl, and All That: An Informal Text on Vector Calculus*, 4th ed., W. W. Norton, 2005.
- [5] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, 1999, ch. 1–4.
- [6] G. Feinberg, “Possibility of faster-than-light particles,” *Physical Review*, vol. 159, no. 5, pp. 1089–1105, 1967.
- [7] Y. Aharonov, A. Komar and L. Susskind, “Superluminal behavior, causality, and instability,” *Physical Review*, vol. 182, no. 5, pp. 1400–1403, 1969.
- [8] P. W. Higgs, “Broken symmetries and the masses of gauge bosons,” *Physical Review Letters*, vol. 13, no. 16, pp. 508–509, 1964.
- [9] S. Coleman, “Fate of the false vacuum: Semiclassical theory,” *Physical Review D*, vol. 15, no. 10, pp. 2929–2936, 1977.
- [10] S. Glasstone and P. J. Dolan, Eds., *The Effects of Nuclear Weapons*, 3rd ed., U.S. Department of Defense and Department of Energy, 1977, ch. 3.
- [11] Jujutsu Kaisen Wiki, *Limitless*, <https://jujutsu-kaisen.fandom.com/wiki/Limitless>, retrieved September 2026.
- [12] Jujutsu Kaisen Wiki, *Cursed Technique Lapse: Blue*, https://jujutsu-kaisen.fandom.com/wiki/Cursed_Technique_Lapse:_Blue, retrieved September 2026.
- [13] Jujutsu Kaisen Wiki, *Cursed Technique Reversal: Red*, https://jujutsu-kaisen.fandom.com/wiki/Cursed_Technique_Reversal:_Red, retrieved September 2026.
- [14] Jujutsu Kaisen Wiki, *Hollow Technique: Purple*, https://jujutsu-kaisen.fandom.com/wiki/Hollow_Technique:_Purple, retrieved September 2026.